

Ablation results: memory and selective agreement

Conditions included: No memory, General KG only, ToM KG only, Full KG

140-agent population, immigration policy topic, 30 simulated days, seed 42, run on a DGX server via vLLM. Each agent exchanges once per simulated hour (100,800 exchanges per condition). Population-level metrics (entropy, left-leaning share, effective clusters) are read at day resolution; acceptance and transition metrics use every individual hourly exchange. See the separate methodology document for how each of these is computed and verified.

Each condition is recapped below before its numbers are shown.

The four conditions

NO MEMORY

No memory. The baseline. Every exchange starts from zero; nothing said in a prior conversation carries forward.

GENERAL KG ONLY

Facts only. Agents remember factual claims raised in conversation (general-dimension triplets), but keep no record of what any specific partner believes.

TOM KG ONLY

Theory of mind only. Agents remember inferred beliefs about a specific partner (tom-dimension triplets), but keep no separate record of general topic facts.

FULL KG

All three dimensions combined: general facts, theory-of-mind beliefs about partners, and the agent's own stated beliefs, extracted and retrieved together.

Summary

Condition	Entropy start -> end	Entropy delta	Left-leaning start -> end	Eff. clusters start -> end
No memory	1.720 -> 1.649	-0.0715	42.1% -> 38.6%	3.04 -> 2.80
General KG only	1.418 -> 1.506	+0.0885	24.3% -> 33.6%	2.20 -> 2.48
ToM KG only	1.586 -> 1.574	-0.0123	42.1% -> 47.1%	2.65 -> 2.65
Full KG	1.675 -> 1.691	+0.0159	40.7% -> 32.1%	2.95 -> 2.92

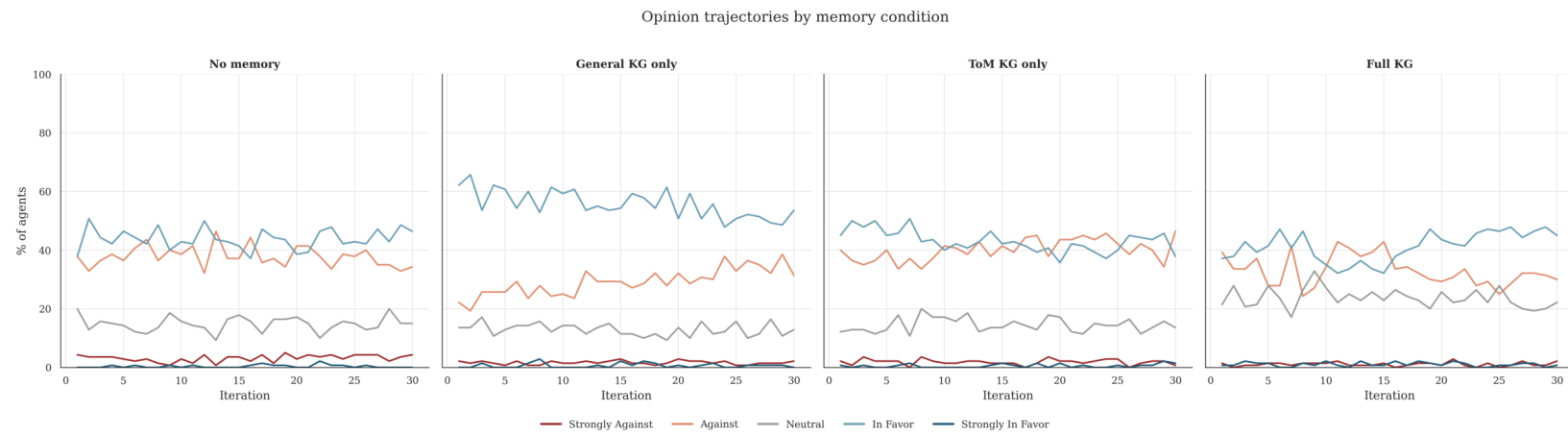
Acceptance rate by stance distance

Distance	No memory	General KG only	ToM KG only	Full KG
-4	95.6%	95.0%	100.0%	95.8%
-3	22.2%	25.0%	26.3%	45.7%
-2	11.2%	10.3%	13.2%	23.6%
-1	23.2%	18.4%	24.9%	31.9%
1	27.4%	36.0%	30.1%	38.6%
2	9.1%	9.6%	9.8%	29.9%
3	55.2%	59.6%	55.7%	57.8%
4	66.7%	80.0%	64.7%	79.2%

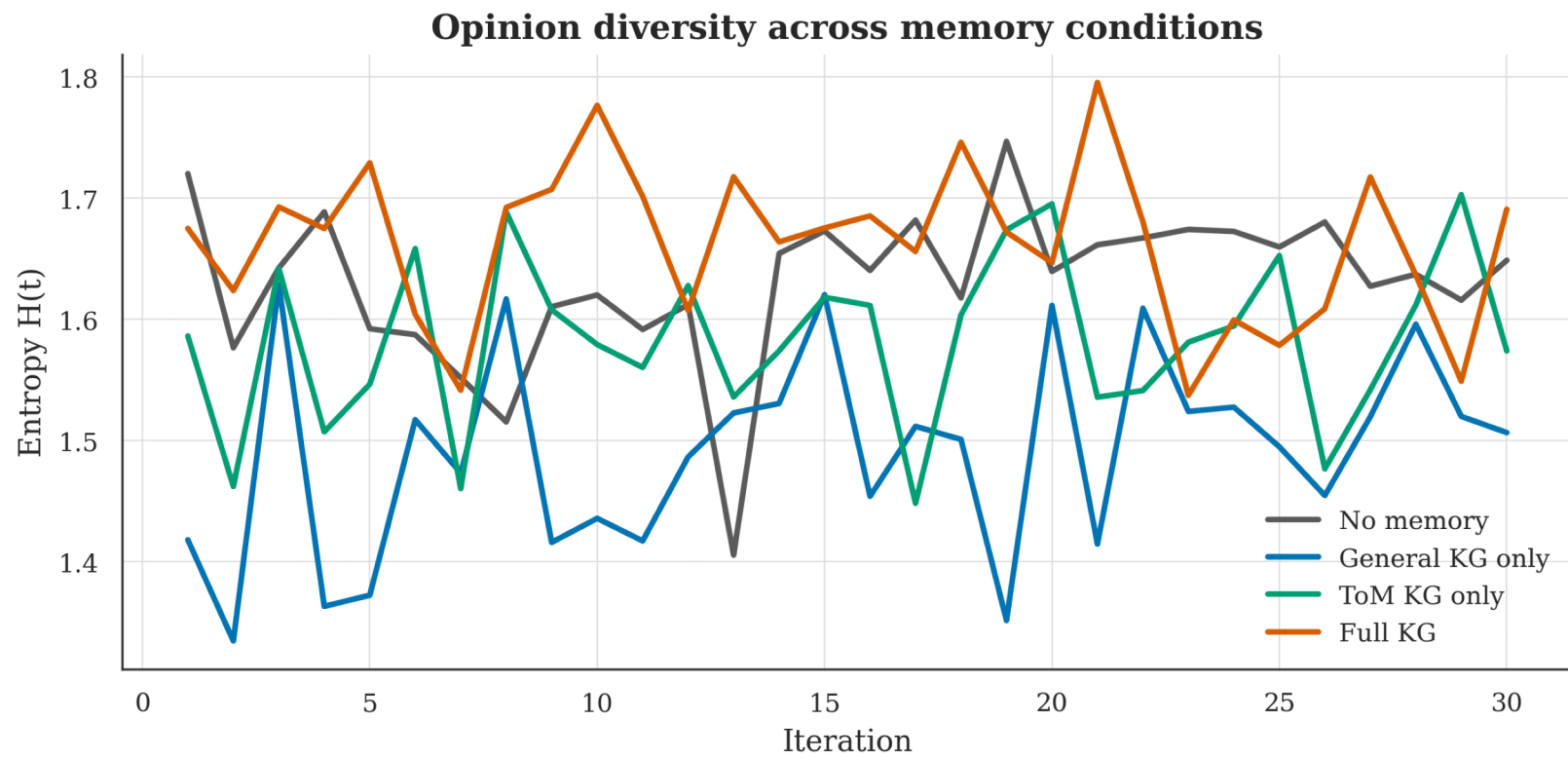
Agent-level stance volatility (fraction of exchanges that changed the agent's stance)

Condition	Median % exchanges resulting in a change	Most stable agent (min % changed)	Least stable agent (max % changed)
No memory	32.7%	Helen Hall (5.3%)	Sasha Kowalski (63.8%)
General KG only	27.4%	Helen Hall (3.0%)	Andrew Singh (48.6%)
ToM KG only	34.2%	Helen Hall (4.8%)	Priya Singh (54.2%)
Full KG	49.0%	Angela Foster (21.3%)	Ralph Cunningham (66.5%)

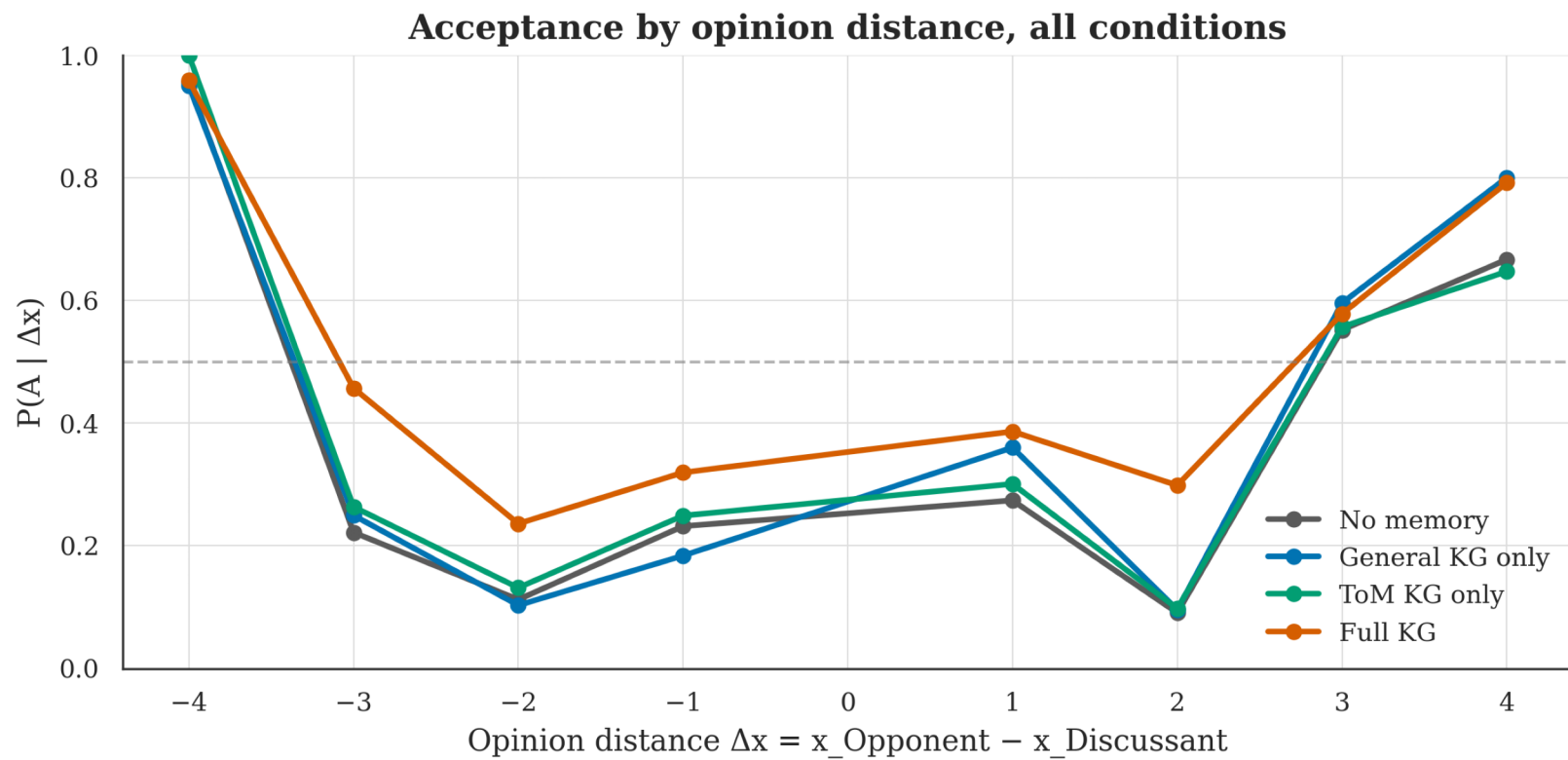
compare_trajectories.png



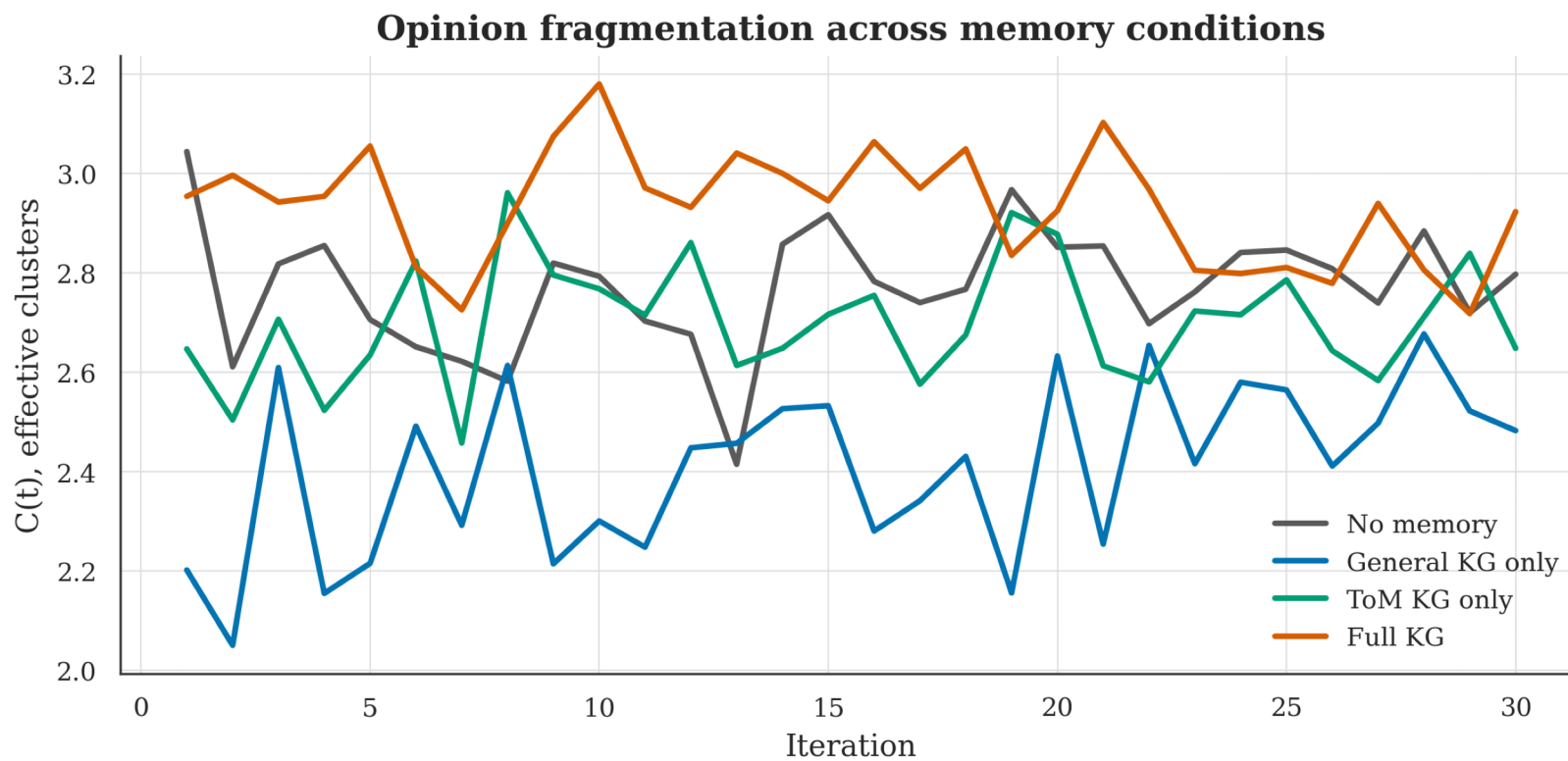
Raw stance distribution over time, one panel per condition. Shows directional drift directly: watch whether one color grows steadily at the expense of the others.



Population diversity over time, one line per condition. A condition that stays higher keeps opinions more spread out; a condition that drops fastest is converging toward consensus fastest.

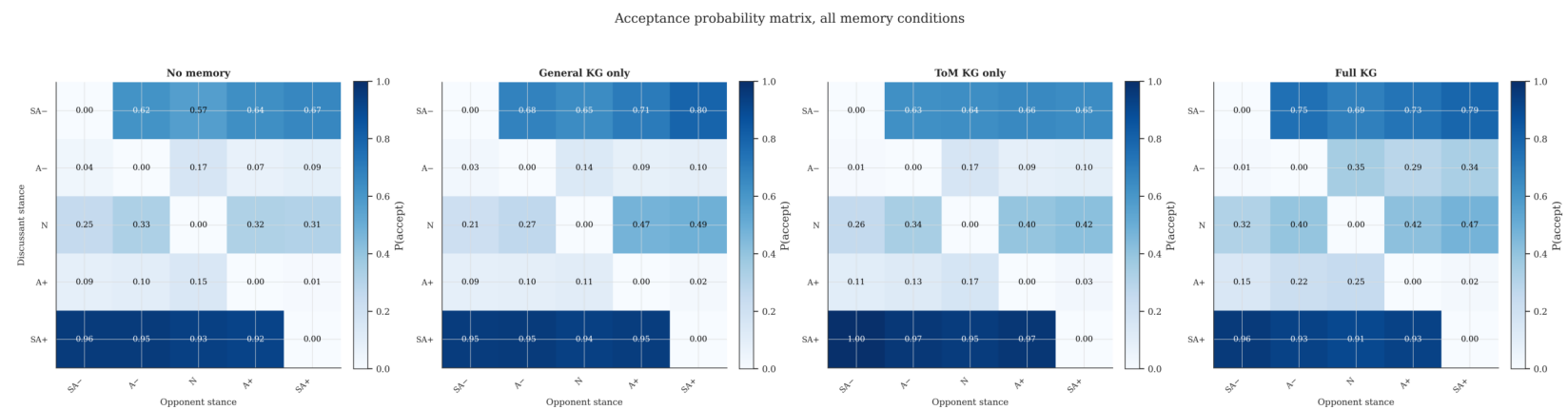


The central test. Each line shows how likely an agent is to shift toward its partner's stance, as a function of how far apart the two started. A flatter line means that condition resists agreement with distant partners more than the baseline does.

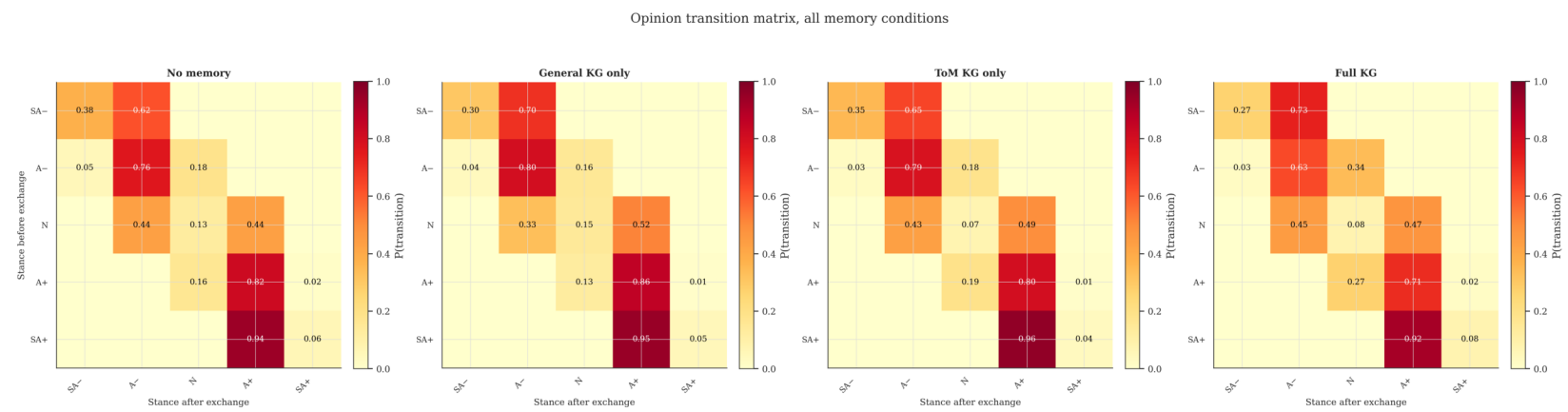


Estimated number of distinct opinion groups still present, per condition, per day. Near 1 means the population converged to one view; near 5 means it stayed spread across the full scale.

compare_acceptance_matrix.png



Full pairwise detail behind the acceptance rate: probability of accepting a shift, broken down by the agent's own stance and the partner's stance, not just the distance between them.



Full pairwise detail on where agents actually ended up: probability of moving from each starting stance to each ending stance, per condition.

Illustrative example: one agent, four conditions

The same persona, agent_064 (Helen Hall), starts at Neutral and ends at In Favor in all four conditions, the identical net outcome. What differs sharply is how she gets there: how often her stated stance actually changes along the way, exchange to exchange.

No memory: 77 of 1440 exchanges changed her stance (5.3%).

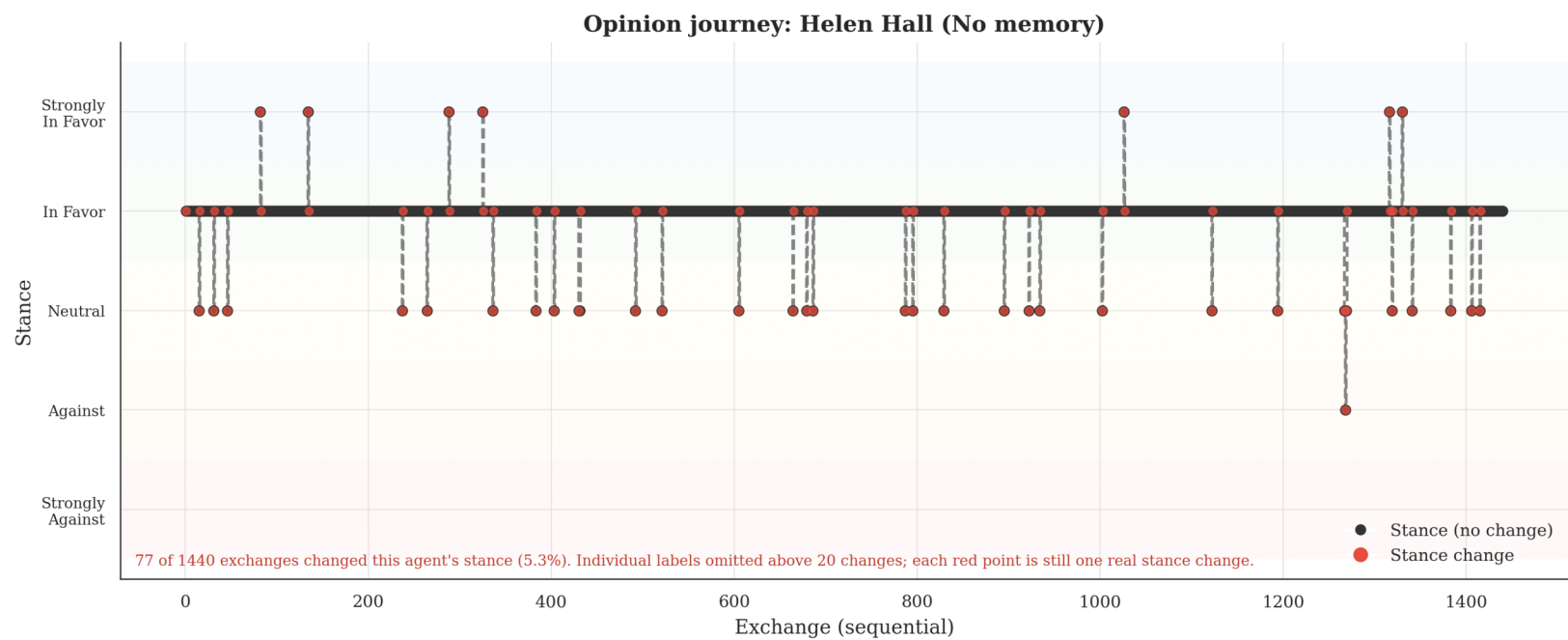
General KG only: 43 of 1440 exchanges changed her stance (3.0%).

ToM KG only: 69 of 1440 exchanges changed her stance (4.8%).

Full KG: 395 of 1440 exchanges changed her stance (27.4%).

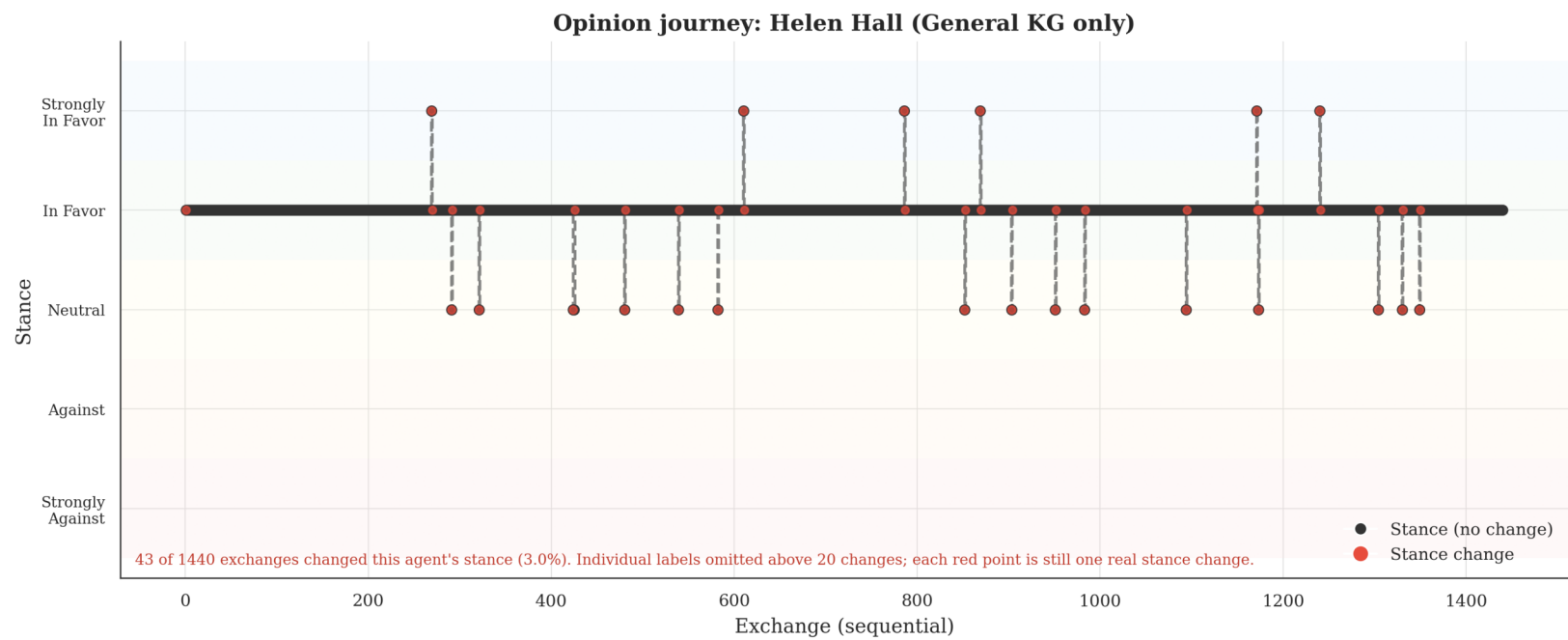
Four journey plots follow, one per condition, same agent.

Agent journey: No memory



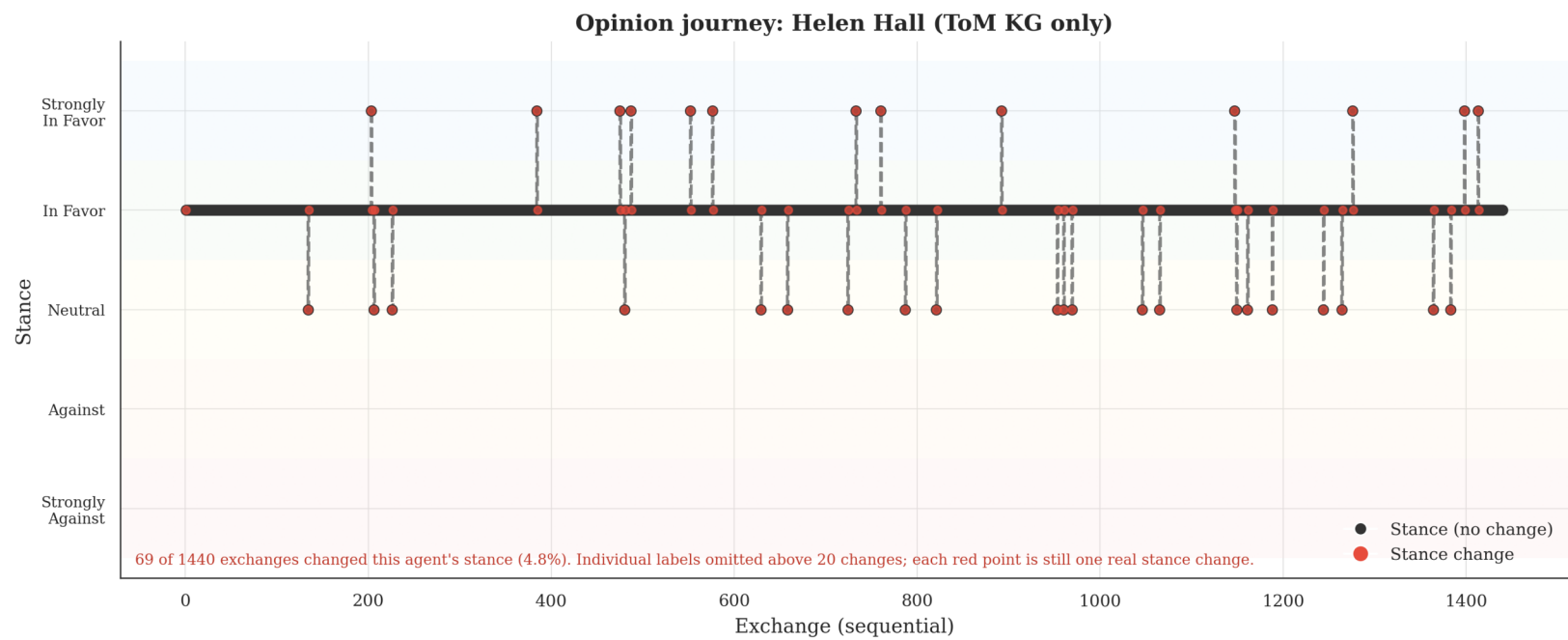
Helen Hall's stance across every exchange in No memory. Red points mark an actual stance change.

Agent journey: General KG only



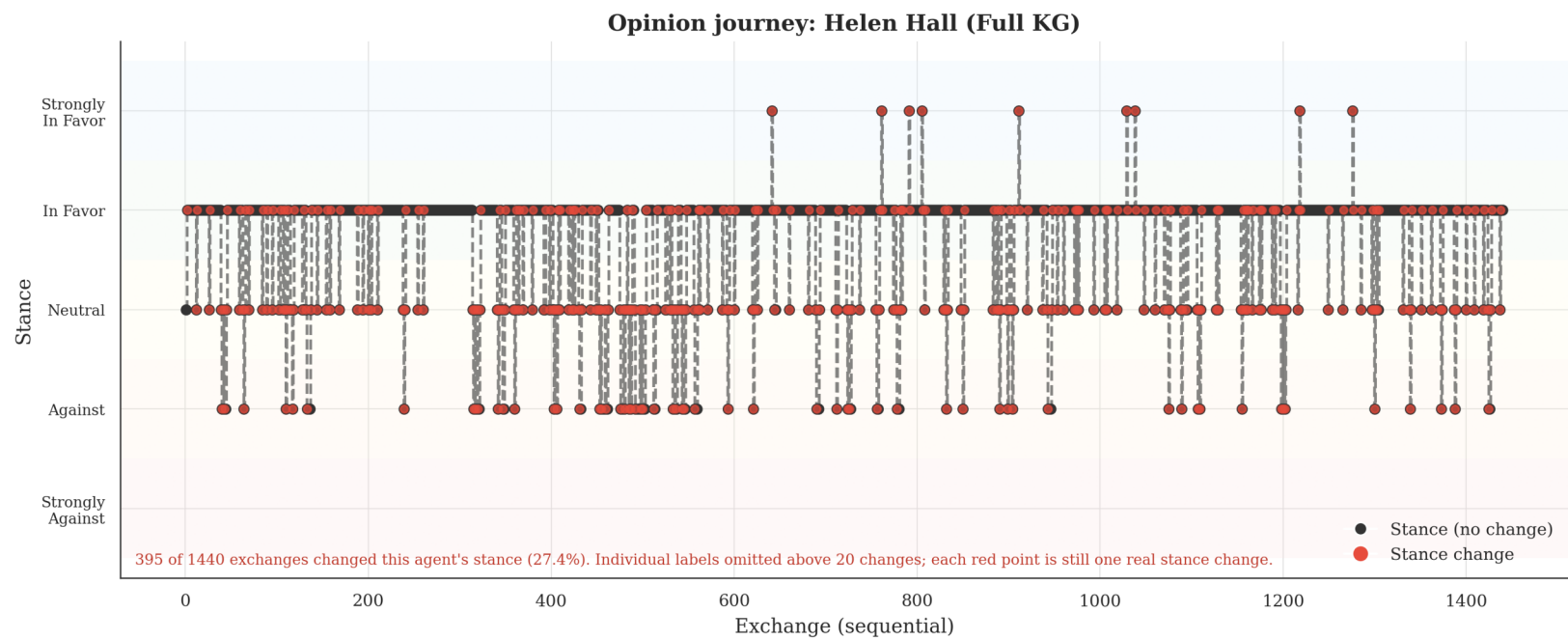
Helen Hall's stance across every exchange in General KG only. Red points mark an actual stance change.

Agent journey: ToM KG only



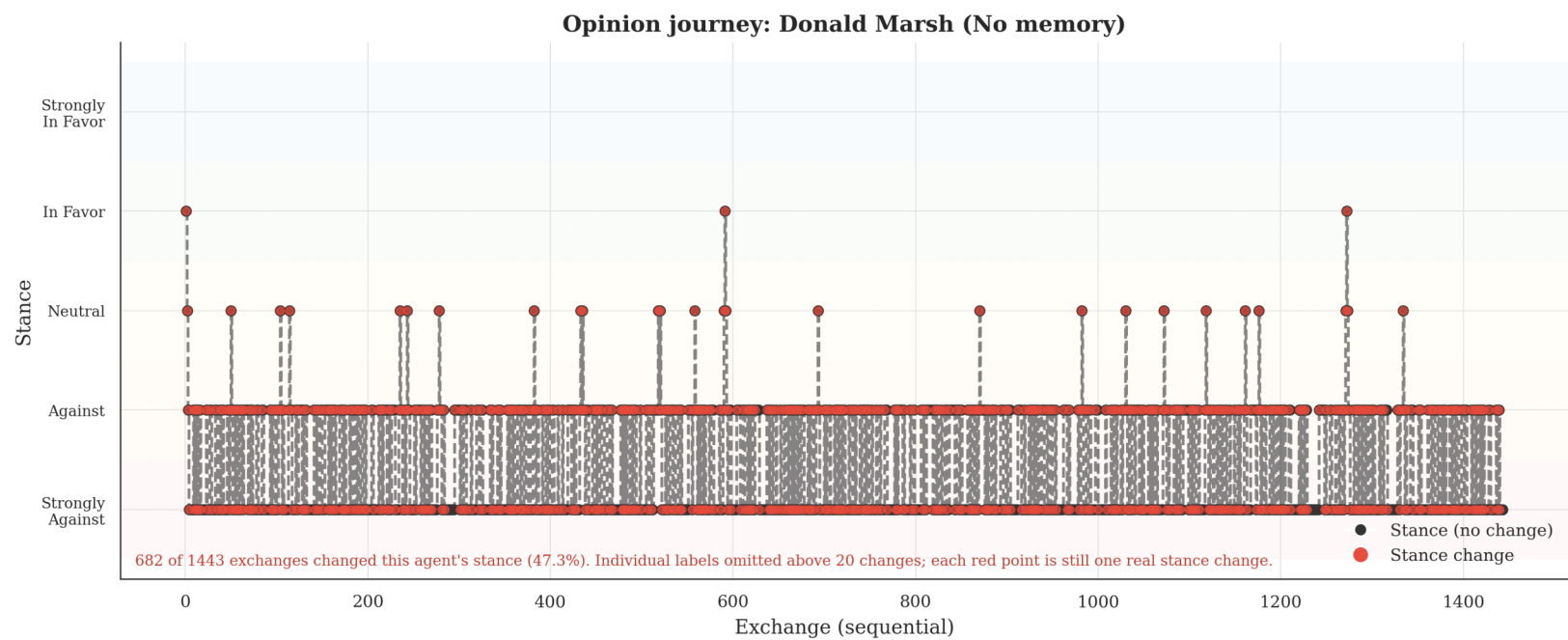
Helen Hall's stance across every exchange in ToM KG only. Red points mark an actual stance change.

Agent journey: Full KG



Helen Hall's stance across every exchange in Full KG. Red points mark an actual stance change.

Largest net mover: no memory



Donald Marsh moved from +2 to -2 net across the run (no memory condition), the largest full-scale reversal observed. 682 of 1443 exchanges changed the stated stance.

What the results say against the hypothesis

Checked against the criteria written down in advance (paper/context/hypothesis_and_scope.md), before these results existed:

1. 'full_kg shows the smallest entropy change over the run':
not confirmed. Smallest |change in entropy| belongs to ToM KG only (0.0123), not full_kg (0.0159). full_kg is the second-most stable by this measure, not the most stable.
2. 'tom_only shows a flatter acceptance-by-distance curve than no_kg and general_only': not confirmed at moderate distance.
At distance 2, tom_only (9.8%) sits between no_kg (9.1%) and general_only (9.6%), not meaningfully flatter. full_kg (29.9%) is the outlier here, well above all three others, the opposite of the predicted direction.
3. 'general_only sits between no_kg and tom_only, or shows no meaningful difference from no_kg': partially confirmed.
general_only's acceptance curve tracks closely with no_kg and tom_only at most distances. Its entropy and left-drift numbers are the most extreme of the three memory conditions, which the original criteria did not anticipate.

The strongest pattern in the data was not one of the three pre-registered criteria. It is agent-level volatility: the fraction of an agent's exchanges that actually change its stated stance.

No memory: median agent changes stance in 32.7% of its exchanges.

General KG only: median agent changes stance in 27.4% of its exchanges.

ToM KG only: median agent changes stance in 34.2% of its exchanges.

Full KG: median agent changes stance in 49.0% of its exchanges.

full_kg agents change their stated stance in roughly half of all

What the results say against the hypothesis (continued)

exchanges, well above the other three conditions, which cluster together. This is the opposite of what the hypothesis predicted for full_kg specifically: combining all three memory dimensions produced the LEAST stable agents observed, not the most stable. The single-agent example on the previous pages (Helen Hall) shows this directly: the same persona, same seed, same starting point, goes from changing its mind on under 5% of exchanges under no_kg/general_only/tom_only to 27.4% under full_kg.

Read plainly: this data does not support the hypothesis as written. It does not refute the idea that memory changes agent behavior, the behavior clearly differs by condition, but the direction is not the one predicted. Combined memory (full_kg) correlates with more agreement and more volatility, not less.

Where to take this next

Suggested talking points for the discussion with Rossetti

1. The exchange cadence changed from once per day to once per hour in the DGX run. Confirm the actual code diff before writing any methodology text describing the loop.
2. The full_kg volatility finding is the headline result and needs a mechanism hypothesis before it goes in a paper. Two candidates worth asking about: does retrieving three dimensions of context per turn (general + tom + beliefs) simply give the model more material to be swayed by, diluting whatever self-consistency signal a single-dimension condition provides? Or is this an FSRS retrieval-weighting artifact specific to having three co-active dimensions competing for retrieval priority?
3. Consider whether this run should be treated as the primary result or as a pilot that motivates a targeted follow-up (e.g. an ablation over the number of active KG dimensions, independent of which ones).
4. The general_only vs no_kg entropy and left-drift gap is larger than expected under the original hypothesis. Worth a specific look at whether fact-only memory has a directional effect the hypothesis did not anticipate.
5. Sample sizes are strong everywhere except the +-4 stance-distance cells (17-45 exchanges per condition). Any claim resting on those cells should be flagged as low-confidence.